Dictionaries

Topics:

- Understanding Dictionaries
- Associated Methods

Dictionary

Dictionaries

Dictionary in Python is an unordered collection of data values, used to store data values like a map, which unlike other Data Types that hold only single value as an element, Dictionary holds **key:value** pair. Key value is provided in the dictionary to make it more optimized.

Creating Dictionaries

In Python, a Dictionary can be created by placing sequence of elements within curly **{}** braces, separated by 'comma'

Examples

```
>>> s = {"name":"vamsi","city":"Hyderabad","gender":"male"}
```

```
>>> print(s)
```

```
{'name': 'vamsi', 'city': 'Hyderabad', 'gender': 'male'}
```

```
>>> print(type(s))
```

```
<class 'dict'>
```

Dictionary Methods

Dictionary Object is defined with the following pre-defined Methods

.clear(...)

Remove all elements from this Dictionary.

.copy()

It creates a shallow copy of the Dictionary object

.get()

It is a conventional method to access a value for a key.

Dictionary Methods

.items()

Returns a list of dict’s (key, value) tuple pairs.

.keys()

Returns list of dictionary dict’s keys

.values()

Returns list of dictionary dict’s values.

Dictionary Methods

.pop()

Removes and returns an element from a dictionary having the given key.

.popitem()

Removes the arbitrary key-value pair from the dictionary and returns it as tuple.

Dictionary Examples

```
>>> x = {"name":"vamsi","city":"hyderbad","gender":"male"}
>>> print(x)
{'name': 'vamsi', 'city': 'hyderbad', 'gender': 'male'}
>>> x.keys()
dict_keys(['name', 'city', 'gender'])
>>> x.values()
dict_values(['vamsi', 'hyderbad', 'male'])
>>> x.items()
dict_items([('name', 'vamsi'), ('city', 'hyderbad'), ('gender', 'male')])
```

Dictionary Examples

```
>>> x.get("name")
'vamsi'
>>> x["city"]
'hyderbad'
>>> x.pop("gender")
'male'
>>> print(x)
{'name': 'vamsi', 'city': 'hyderbad'}
>>> x.popitem()
('city', 'hyderbad')
>>> print(x)
{'name': 'vamsi'}
```

Dictionary Examples

```
>>> print(x)
{'name': 'vamsi'}
>>> x["country"]="India"
>>> print(x)
{'name': 'vamsi', 'country': 'India'}
>>> x.clear()
>>> print(x)
{}
```